MILITARY SPECIFICATION

EPOXY RESIN SYSTEMS FOR CONCRETE REPAIR

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers epoxy resin systems for use in sealing, bonding, grouting, and repairing of portland cement concrete and in bonding and grouting of associated equipment, such as bolts, valves, and metal bars.

1.2 Classification.

1.2.1 Composition, types, and grades. Epoxy resin systems shall be of the following compositions, types, and grades as specified (see 6.2):

Composition F - Filled epoxy.

Type I - For use in bonding hardened concrete and other materials to hardened concrete.

Grade 2 - Medium viscosity.

Grade 3 - Non sagging consistency.

Type III - For use as a binder in epoxy mortars or epoxy concretes.

Grade 1 - Low viscosity.

Grade 2 - Medium viscosity.

Grade 3 - Non sagging consistency.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5610

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

AMSC N/A

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:27Z
Check the source to verify that this is the current version before use.

Composition U - Unfilled epoxy.

Type I - For use in bonding hardened concrete and other materials to hardened concrete.

Grade 1 - Low viscosity.

Type III - For use as a binder in epoxy mortars or epoxy concretes.

Grade 1 - Low viscosity.

1.2.2 Classes. Epoxy resin systems shall be of the following classes as specified (see 6.2).

Class B - For use when materials, substrate, and atmospheric temperatures are between 40 degrees Fahrenheit (°F) and 60°F (4.5° Celsius [C] and 15.6°C).

Class C - For use when materials, substrate, and atmospheric temperatures are between 60°F and 100°F (15.6°C and 37.8°C).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

PPP-C-2020 - Chemicals, Liquid, Dry and Paste, Packaging of.

STANDARDS

FEDERAL

FED-STD-313 - Materials Safety Data Sheets, Preparation and Submission of.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-147 - Palletized Unit Loads.

(Copies of standards and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:27Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERAILS (ASTM)

C881 - Epoxy-Resin-Base Bonding Systems for Concrete.
C882 - Bond Strength of Epoxy-Resin Systems used with Concrete.
C883 - Effective Shrinkage of Epoxy-Resin Systems used with Concrete.
C884 - Thermal Compatibility between Concrete and an Epoxy-Resin Overlay.
D570 - Test Method for Water Absorption of Plastics.
D1259 - Test Methods for Nonvolatile Content of Resin Solutions.
D1763 - Standard Specification for Epoxy Resins.
D2393 - Test Method for Viscosity of Epoxy Resins and Related Compounds.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be as specified herein and in applicable specifications and standards, and other referenced documents. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free of defects which adversely affect performance or serviceability of the finished product.

3.2 Epoxy resin systems. The epoxy resin systems shall be two-component systems for combining immediately prior to use and shall consist of the epoxy resin as component A and the curing agent as component B. Suitable flexibilizers may be included to provide necessary characteristics. Volatile solvent additives shall not be used. Suitable inert fillers shall be combined in one or both components of the composition F systems. During the curing process the resultant polymer shall not be adversely affected by moisture

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:27Z
Check the source to verify that this is the current version before use.

present on damp surfaces, but are not formulated for underwater applications. The epoxy systems specified herein do not include emulsive epoxies applicable to mixing with wet unhardened portland cement concrete to form epoxy modified concrete.

3.2.1 Epoxy resin. The epoxy resin in component A shall conform with ASTM D1763, type I, grade 1, class A.

3.2.2 Flexibilizer. The flexibilizer, when used, shall be contained in component A or in component B and shall be compatible with all other constituents in the system.

3.2.3 Curing agent. The curing agent shall be furnished in component B in sufficient proportions to polymerize the resin into a solid in accordance with the class specified. The curing agent shall be one compound or a combination of curing agents as needed to meet the requirements of this specification.

3.2.4 Filler. Composition F systems shall contain fillers combined into component A or component B, or both, as needed to meet the requirements for the type and grade specified. Composition U systems shall not contain fillers. Filler is defined as a non-settling mineral filler, thixotropic agent, or pigment. The filler shall be factory mixed into the components using high speed dispersion equipment or other means to insure thorough mixing. The quantities and types of fillers used, and factory mixing, shall provide components which can be easily handled and mixed in the field. Sufficient thixotropic agent shall be used for those systems required to have thixotropic properties. Pigment used shall provide, after mixing and curing, a solid product that is gray in color similar to that of hardened portland cement concrete.

3.3 Physical requirements. When mixed as recommended by the contractor, the epoxy systems shall conform to the physical requirements described in Table I.

3.4 Gel time (pot life). When specified (see 6.2), the gel time requirement specified in Table I shall be waived when automatic mixing and dispensing equipment is to be used.

3.5 Storage stability. The epoxy systems shall be capable of meeting the requirements specified in 3.3 after the components are stored for one year as specified in 4.5.9.

3.6 Marking requirements.

3.6.1 Identification marking. Identification shall be permanently and legibly marked directly on each container of each component at the source of manufacture. Identification shall include the specification number, applicable composition, type, grade and class, as well as the component designation, date of manufacture, shelf life, the manufacturer's name and address, and manufacturer's product identifier.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:27Z
Check the source to verify that this is the current version before use.

3.6.2 Handling, mixing, and storage instructions. Complete instructions shall be placed on the outside of all containers of each component by stencil, lithograph, or securely affixed labels which shall include the following:

a. Mixing ratio of each component by volume, mixing method, and mixing time.
b. Handling precautions.
c. Storage conditions.

3.7 Material Safety Data Sheets (MSDS). MSDS shall be prepared in accordance with FED-STD-313 and submitted as directed in the contract or order at the time of acquisition award. Copies shall be forwarded to the designated Industrial Hygienist and the focal point of the activity that purchased the item and the focal point of the using activity if different (see 6.2, 6.3, and 6.4).

3.8 Workmanship. The workmanship shall be such as to insure a product which is uniform and free of contamination and whose properties are in accordance with this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that supplies and materials are inspected for compliance with all the requirements specified herein and in applicable referenced documents.

4.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one epoxy system kit consisting of the epoxy resin component and the curing agent component. All kits of the same classification, offered for delivery at one time shall be considered a lot for the purpose of inspection. The inspection level shall be level I and the Acceptable Quality Level shall be 2.5 percent defective. If an inspection lot is rejected, when so approved by the contracting officer the contractor may correct the defects, or screen out the defective units, and resubmit for a complete reinspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separated from new lots, and shall be clearly identified as reinspected lots.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:27Z
Check the source to verify that this is the current version before use.

4.4 Examination. Each sample kit shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations for defects, including the following. Noncompliance with any specified requirements shall constitute one defect.

101. Mixing and handling instructions and marking not as specified, missing, or illegible.
102. Containers leaking or otherwise damaged.
103. Uneven distribution of materials in containers, or settling out of fillers.
104. Color of composition F epoxy system not as specified after mixing.
105. Missing components.

4.5 Tests. Each sample production unit shall be subjected to the tests of 4.5.1 through 4.5.9. Test for the properties in Table I and 3.5. Failure to pass any test shall constitute cause for rejection.

4.5.1 Viscosity Viscosity shall be determined in accordance with ASTM D2393 except that temperatures for the determination shall be 50 ± 1.8°F (10 ± 1°C) for class B and 73 ± 1.8°F (23 ± 1°C) for class C.

4.5.2 Consistency. Consistency of the freshly mixed epoxy system shall be determined in accordance with ASTM C881.

4.5.3 Bond strength. Bond strength of the cured epoxy system shall be determined in accordance with ASTM C882.

4.5.4 Shrinkage. The effective shrinkage of the epoxy system during the curing process shall be determined in accordance with ASTM C883, except that the thickness of the layer of epoxy shall be as shown in Table I for the applicable types.

4.5.5 Thermal compatibility. Thermal compatibility between the concrete substrate and the epoxy system shall be determined in accordance with ASTM C884 except that the thickness of the layer shall be 2.0 inches (50 millimeters (mm)) and the proportion of sand/aggregate to epoxy shall be as recommended by the manufacturer or formulator of the epoxy resin systems.

4.5.6 Volatile content. Volatile content of component A and of the mixed epoxy polymer shall be determined in accordance with method B of ASTM D1259.

4.5.7 Gel time. Gel time of the freshly mixed epoxy system shall be determined in accordance with ASTM C881.

4.5.8 Absorption. Moisture absorption of the cured epoxy system shall be determined in accordance with ASTM D570 with test modifications as cited in ASTM C881.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:27Z
Check the source to verify that this is the current version before use.

4.5.9 Storage stability test. Separate components A and B shall be stored in original unopened one gallon containers at 75°F ± 9°F (24°C ± 5°C) for one year after date of manufacture. The components shall then be mixed according to the manufacturer's instructions and the epoxy mixture subjected to the tests specified herein to determine conformance to the requirements specified in 3.5. The contractor shall make available to the contracting officer or his authorized representative evidence of compliance with this test, including a report of actual test results indicating that this test has been performed on systems previously manufactured from the same materials and proportions as those used in the systems being offered for delivery.

4.6 Packaging inspection. The inspection of the preservation, packing, palletization, and marking shall be in accordance with the requirements of section 4 of PPP-C-2020.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with requirements of PPP-C-2020 with the level of preservation and the level of packing as specified (see 6.2).

5.2 Palletization. Material shall be palletized in accordance with MIL-STD-147 when the following criteria are met:

a. Load to consist of four or more unskidded containers.
b. Load shall utilize a minimum of 80 percent of the pallet base.

6. NOTES

6.1 Intended use.

6.1.1 Composition F, Type I, grades 2 and 3. Composition F, Type I, grades 2 and 3 should be selected for non-pressure grouting of cracks and for cementing dowels.

6.1.2 Composition F, Type III, grades 1, 2, and 3. Composition F, Type III, grades 1, 2, and 3 should be selected for repair of vertical walls and overhead surfaces.

6.1.3 Composition U, Type I, grade 1. Composition U, Type I, grade 1 should be selected for pressure grouting of cracks.

6.1.4 Composition U, Type III, grade 1. Composition U, Type I, grade 1 should be selected for repair of portland cement concrete pavement.

6.1.5 Classes B and C. Classes B and C should be selected as appropriate for the temperature ranges anticipated in field applications.

6.2 Ordering data.

a. Title, number, and date of this specification.
b. Composition, type, and grade required (see 1.2.1).
c. Class required (see 1.2.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:27Z
Check the source to verify that this is the current version before use.

d. When minimum gel time of 30 minutes is not required (see 3.4).
e. Addresses for submission of MSDS (see 3.7 and 6.4).
f. Level of preservation and level of packing required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of DoD Federal Acquisition Regulations (FAR) Supplement 27.410-6, the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DoD FAR 27.410-6 are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 MSDS submission and forwarding. After review and acceptance of MSDS by designated recipients, approved copies will be forwarded to arrive at destinations prior to material delivery (see 3.7).

6.5 Definition of terms.

6.5.1 Binder: The resin system used in an epoxy mortar or concrete that binds the aggregate into a cohesive mass.

6.5.2 Component: One of the two parts of the epoxy resin system containing either the resin or curing agent which, when mixed together, form a solid polymer.

6.5.3 Curing agent: The constituent that causes the conversion of a fluid epoxy to a solid polymer by means of a chemical reaction.

6.5.4 Epoxy concrete: A combination of epoxy binder and fine/course aggregate in a consistency similar to portland cement concrete.

6.5.5 Epoxy grout: An epoxy system thickened by means of fillers, used to fill cracks, set dowels and anchor bolts, etc., similar to conventional grout.

6.5.6 Epoxy mortar: A combination of epoxy binder and fine aggregate in a consistency suitable for troweling.

6.5.7 Filler: A relatively inert material incorporated in the epoxy system to increase the hardness and strength, decrease expansion and contraction, change the color or consistency, or lower the cost.

6.5.8 Flexibilizer: A substance which reacts with epoxy formulations to impart flexibility, usually at the cost of a small decrease in adhesion.

6.5.9 Gel time (pot life). The time after mixing that the epoxy systems shall remain workable, spreadable, and of smooth consistency.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T22:27Z
Check the source to verify that this is the current version before use.

6.6 Correlation with ASTM specifications. The types, grades, and classes of epoxy systems used in this specification (see 1.2.1 and 1.2.2) are comparable to the types, grades, and classes set forth in ASTM C881 except that type II and class A are omitted and class C has an upper limit of 100°F in this specification.

Preparing Activity:

Navy - YD

(Project 5610-N269)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:27Z
Check the source to verify that this is the current version before use.

Table I. Physical requirements of epoxy systems.

Requirement	Composition F					Composition U		Test Paragraph
	Type I		Type III			Type I	Type III	
	Grade 2	Grade 3	Grade 1	Grade 2	Grade 3	Grade 1	Grade 1	
Viscosity, poise, Pascal second (Pa.s)	20(2.0)min 100(10.0)max	-- --	.7(.07)min 20(2.0)max	20(2.0)min 100(10.0)max	-- --	.3(.03)min 20(2.0)max	.3(.03)min 20(2.0)max	4.5.1
Consistency, inches sag, max, millimeter (mm)	--	1/4 (6.4)	--	--	1/4 (6.4)	--	--	4.5.2
Bond strength, psi, megapascal (MPA) 2 days dry cure 14 days moist cure	2000(13.8) 2000(13.8)	2000(13.8) 2000(13.8)	300(2.06) 300(2.06)	300(2.06) 300(2.06)	300(2.06) 300(2.06)	2000(13.8) 2000(13.8)	300(2.06) 300(2.06)	4.5.3
Shrinkage 1/16-inch (1.6mm) layer 1/8-inch (3.2mm) layer	pass test --	pass test --	-- pass test	-- pass test	-- pass test	pass test --	-- pass test	4.5.4
Thermal compatibility	--	--	pass test	pass test	pass test	--	pass test	4.5.5
Volatile content, max %	3	3	--	--	--	3	--	4.5.6
Gel time, minutes minimum maximum	30 - all classifications 120 - all classifications							4.5.7
Absorption, % maximum	1.5 - all classifications							4.5.8

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:27Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

Commanding Officer (156)
Naval Construction Battalion Center
Port Hueneme, CA 93043-5000

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer (156)
Naval Construction Battalion Center
Port Hueneme, CA 93043-5000

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:27Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-29245(YD)	EPOXY RESIN SYSTEMS FOR CONCRETE REPAIR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:27Z
Check the source to verify that this is the current version before use.